

VIKAS PAL

Aspiring Software Engineer | AI/ML & Generative AI Enthusiast | Open Source Advocate | LLD Learner
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

JNTUH University College of Engineering Sultanpur

Bachelor of Technology (B.Tech), Computer Science and Engineering | CGPA: 8.08

Sultanpur, Telangana

2023 - Expected 2026

Quli Qutub Shah Govt Polytechnic College

Diploma in Computer Science and Engineering | CGPA: 9.37 (Gold Medalist)

Hyderabad, Telangana

2020 - 2023

EXPERIENCE

Research Intern – Machine Learning & Secure Code Analysis (under Prof. Jyothi Vedurada) | IIT

Hyderabad, Compilers Lab

Hyderabad, February 2025 – May 2025

- Engineered machine learning models using graph kernels to enhance API selection for intelligent code suggestions, boosting developer productivity. Led research on software vulnerability detection, creating robust encoding schemes to identify code-level security weaknesses effectively.

Robotics Research Intern (under Prof. C Krishna Mohan) | IIT Hyderabad

Hyderabad, September 2024 – March 2025

- Engineered and optimized an autonomous navigation system for Unmanned Ground Vehicles (UGVs) by integrating A* pathfinding with SLAM (Simultaneous Localization and Mapping) algorithms, enabling robust real-time obstacle avoidance in GPS-denied and unstructured terrains; reduced route planning latency by 30

Contributor | Google Summer of Code (GSoC) – AboutCode Organization

Remote, January 2023 – November 2023

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- Contributed to open-source development under AboutCode by enhancing UI responsiveness and accessibility with React, optimizing backend API endpoints for improved performance, and collaborating with mentors and global contributors through GitHub and weekly Agile sprints to ensure maintainable, well-documented code.

Industrial Training | Java & Spring Boot Developer

Hyderabad, June 2022 – November 2022

- Engineered RESTful APIs and microservices using Spring Boot, Hibernate, and Docker with secure JWT-based authentication, while designing and implementing CI/CD pipelines using Jenkins, Docker, and Kubernetes for automated deployments.

SKILLS

Programming Languages: C, C++, Java, Python, JavaScript

Libraries/Frameworks: React.js, Spring Boot, Express.js, NumPy, Pandas, TensorFlow, Keras, OpenCV, YOLOv5

Tools/Platforms: Git, GitHub, VS Code, Docker, Kubernetes, Jenkins, Apache Kafka, Postman, Figma

Linux

Databases:

SQL, MongoDB

PROJECTS

LLM-Based Automated Script Generator | [Link](#)

GPT-2, LLaMA, Gradio, Hugging Face, Fine Tuning

- Engineered a machine learning-based script generation solution using Transformer models (GPT-2, LLaMA), fine-tuned on a custom dataset to enhance contextual accuracy. Deployed the system on Hugging Face Spaces, enabling real-time interaction and wide public accessibility, while optimizing for performance and scalability.

GeoSecure: Emergency Alert & Tracking App | [Link](#)

Java, WebSocket, Apache Kafka

- Engineered a real-time emergency alert system with geolocation tracking, WebSocket communication, and Apache Kafka for secure and reliable message streaming across distributed systems.

Tumor Detection in MRI Scans with YOLOv8 | [Link](#)

Python, YOLO

- Engineered and trained a YOLO-based model for brain tumor detection in medical images, achieving over 90% accuracy. Applied advanced preprocessing techniques, such as normalization and data augmentation, to enhance model performance and generalization.

DevMatch - Intelligent Developer Collaboration Platform | [Link](#)

Spring Boot, React, JWT, MongoDB,

WebSocket, SOLID Principles

- Engineered a real-time full-stack web application with secure authentication, AI-driven skill and project compatibility algorithms, real-time messaging, and advanced search features — boosting developer match success by 60%.

CERTIFICATIONS

• Java Foundation - **Oracle**

• Problem Solving - **HackerRank**

• AI/ML for Geodata Analysis - **ISRO**

• Digital Image Processing - **NPTEL**

ACHIEVEMENTS

- Selected for a prestigious one-year Diploma Apprenticeship in Computer Science at DRDO – CHESS (Centre for High Energy Systems & Sciences), Ministry of Defence, Hyderabad, under the supervision of Scientist 'F' Mr. Sanjeev Kumar.
- Awarded First Prize for innovative project at the state-level Project Competition at Surjana Tech Fest.
- Technical Coordinator for university-level hackathons and technical workshops.
- Coding Profiles: LeetCode (leetcode.com/u/DDP4LBBXnJ/), GeeksforGeeks ([geeksforgeeks.org/user/vikas83pal](https://www.geeksforgeeks.org/user/vikas83pal))
- Published a journal article on women's safety solutions titled "*Smart Emergency Response System for Women Safety*" in IJERT. Available at: [ijert.org/...](https://www.ijert.org/)